

Group Project: Causal Inference Paper Exposition

Quantitative Methods for Humanities

2026–27

Purpose

The goal of this project is to show that your group can read, understand, and explain a published paper that uses causal inference. You are not being asked to carry out a new causal analysis in R or to work with the assigned course datasets. Instead, your task is to study one paper carefully and give a clear, brief exposition of its research question, evidence, causal strategy, results, and limitations.

The project is designed to reward understanding rather than technical decoration. A good presentation should make the paper understandable to classmates who have taken this course but have not read the paper.

Basic Rules

- Groups may have at most **three students**.
- The project is worth **10%** of the final grade.
- Each group will choose and analyze one **causal-inference paper** from the list of papers provided by the instructor.
- One group member should submit the slides and the short report as **PDF files** on Canvas.
- The deliverables are:
 - a short oral presentation, supported by clear and informative slides,
 - a short report introducing and explaining the paper.
- Every member of the group must understand the whole paper. The instructor may ask any member to answer any question.

Paper Selection

Each group will either choose or be assigned a paper from the list provided in the Papers section at the end of this document.

Proposing your own paper

If your group wants to propose another paper, ask for approval before working on it. A suitable paper should:

- study a social, political, cultural, historical, educational, economic, or policy question;
- make an explicit causal claim or evaluate a causal effect;
- identify a treatment, intervention, exposure, reform, event, or assignment mechanism;
- define at least one outcome variable;
- use evidence that can be explained with the causal inference concepts covered in class.

Presentation Format

The presentation should be at most **10 minutes** (longer presentations will be penalized), followed by a short round of questions. As a guideline, use no more than **6 content slides**, plus one final slide for references if needed.

Slides should be clear and selective. Language should be precise and straightforward, not vague or ambiguous. Do not try to summarize every detail of the paper. Focus on the core causal argument and the evidence needed to understand it.

Short Report Format

The short report should follow the same structure as the presentation. It must be at most **5 pages**, excluding references, and use a readable format: **12 pt font**, at least **2.5 cm margins**, and **1.5 line spacing**. Do not use dense formatting to fit more text.

Required Content

Your presentation and report should cover the following points. Some papers may not contain all of them in the same way. If something is not applicable, say so briefly and explain why.

1. Paper, Problem, and Motivation

Begin with the full citation of the paper. Explain:

- what problem the paper studies;
- why the problem matters;
- what gap, debate, or puzzle the paper tries to address.

2. Research Question and Main Claim

State the main research question in your own words. If possible, write it as a causal question. Then state the paper's main answer or claim.

Be precise. Avoid vague descriptions such as “the paper studies inequality” or “the paper talks about education.” Instead, identify what is being compared and what outcome is being measured.

3. Causal Framework

Identify the main causal elements of the paper:

- the unit of analysis;
- the treatment, intervention, or exposure;

- the outcome variable or variables;
- the comparison or control group;
- the causal effect or estimand, when the paper defines one;
- the counterfactual idea behind the paper’s argument.

If the paper has several treatments or outcomes, focus on the most important one and mention the others only if needed and time allows.

4. Data and Descriptive Evidence

Explain the data used in the paper:

- the data source;
- the sample and time period;
- the key variables;
- any descriptive table, figure, or exploratory evidence that helps motivate the analysis.

You may reproduce or adapt one figure or table from the paper if it is central to the argument. If you do, cite the original table, figure, or page clearly.

5. Causal Inference Technique

Explain the paper’s identification strategy in words. Possible techniques include randomized experiments, natural experiments, instrumental variables, regression discontinuity, difference-in-differences, matching, panel fixed effects, or another design.

Your explanation should answer:

- Why do the authors think their comparison can be interpreted causally?
- What assumption is most important for the causal interpretation?
- What could threaten that assumption?
- What checks, placebo tests, balance tests, robustness checks, or sensitivity analyses do the authors use, if any?

Use and relate to course concepts when relevant: treatment, outcome, control group, potential outcomes, selection bias, confounding, randomization, parallel trends, average treatment effect, internal and external validity. If needed to understand the paper, you must research other concepts not explored in the course.

6. Results, Interpretation, and Uncertainty

Present the main result. Explain the direction, size, and meaning of the estimated effect in plain language.

If the paper reports confidence intervals, standard errors, p -values, or hypothesis tests, explain what they add to the interpretation. Do not simply say that a result is “significant.” Explain what hypothesis is being tested and what the evidence suggests.

7. Conclusions and Limitations

End by explaining:

- what the authors conclude;

- whether you think the causal interpretation is convincing;
- the strongest limitation of the paper;
- what you learned about causal inference and its applications from reading the paper.

This section should include your group’s judgment, not only a summary of the authors’ conclusion.

Submission Checklist

Before submitting, check that your slides and report include:

- the names of all group members;
- the full citation of the paper;
- the paper’s research question and main claim;
- treatment, outcome, comparison group, and causal method;
- the main data source and one key table, figure, or result;
- a careful interpretation and one limitation;
- references to the relevant paper pages, tables, or figures;

Papers

1. Salinas, Paula, and Albert Solé-Ollé. 2018. “Partial Fiscal Decentralization Reforms and Educational Outcomes: A Difference-in-Differences Analysis for Spain.” *Journal of Urban Economics* 107: 31–46. doi: 10.1016/j.jue.2018.08.003.
2. Felfe, Christina, Natalia Nollenberger, and Núria Rodríguez-Planas. 2015. “Can’t Buy Mommy’s Love? Universal Childcare and Children’s Long-Term Cognitive Development.” *Journal of Population Economics* 28(2): 393–422. doi: 10.1007/s00148-014-0532-x.
3. Brassiolo, Pablo. 2016. “Domestic Violence and Divorce Law: When Divorce Threats Become Credible.” *Journal of Labor Economics* 34(2): 443–477. doi: 10.1086/683666.
4. Fernández-Kranz, Daniel, and Núria Rodríguez-Planas. 2021. “Too Family Friendly? The Consequences of Parent Part-Time Working Rights.” *Journal of Public Economics* 197: 104407. doi: 10.1016/j.jpubeco.2021.104407.
5. García-Hombrados, Jorge, Marta Martínez-Matute, and Carmen Villa Llera. 2024. “Specialised Courts and the Reporting of Intimate Partner Violence: Evidence from Spain.” *Journal of Public Economics* 239: 105243. doi: 10.1016/j.jpubeco.2024.105243.
6. Torun, Huzeyfe. 2019. “Ex-Ante Labor Market Effects of Compulsory Military Service.” *Journal of Comparative Economics* 47(1): 90–110. doi: 10.1016/j.jce.2018.12.001.
7. Gorjón, Lucía, David Martínez de Lafuente, and Gonzalo Romero. 2024. “Employment Effects of the Minimum Wage: Evidence from the Spanish 2019 Reform.” *SERIEs: Journal of the Spanish Economic Association* 15: 1–55. doi: 10.1007/s13209-023-00291-1.
8. Arranz, José M., and Carlos García-Serrano. 2025. “Assessing the Impact of an Increase in the Minimum Wage on Household Income and Poverty.” *Social Science Research* 127: 103143. doi: 10.1016/j.ssresearch.2025.103143.
9. García-Vega, María, Richard Kneller, and Joel Stiebale. 2021. “Labor Market Reform and Innovation: Evidence from Spain.” *Research Policy* 50(5): 104213. doi: 10.1016/j.respol.2021.104213.
10. González-Val, Rafael, and Miriam Marcén. 2024. “Population Growth and Taxes: The Effect of Regional Differences in the Spanish Inheritance Tax.” *The Annals of Regional Science* 73:

- 135–163. doi: 10.1007/s00168-023-01253-y.
11. Yoshimura, Yuji, Yusuke Kumakoshi, Yichun Fan, Sebastiano Milardo, Hideki Koizumi, Paolo Santi, Juan Murillo Arias, Siqi Zheng, and Carlo Ratti. 2022. “Street Pedestrianization in Urban Districts: Economic Impacts in Spanish Cities.” *Cities* 120: 103468. doi: 10.1016/j.cities.2021.103468.
 12. Valdés, Manuel T., Mar C. Espadafor, and Risto Conte Keivabu. 2025. “Can a Low Emission Zone Improve Academic Performance? Evidence from a Natural Experiment in the City of Madrid.” *Population and Environment* 47: 8. doi: 10.1007/s11111-025-00477-8.
 13. Pinilla, Jaime, Beatriz G. López-Valcárcel, and Miguel A. Negrín. 2019. “Impact of the Spanish Smoke-Free Laws on Cigarette Sales, 2000–2015: Partial Bans on Smoking in Public Places Failed and Only a Total Tobacco Ban Worked.” *Health Economics, Policy and Law* 14(4): 536–552. doi: 10.1017/S1744133118000270.
 14. Lidón-Moyano, Cristina, Marcela Fu, Montse Ballbè, Juan Carlos Martín-Sánchez, Nuria Matilla-Santander, Cristina Martínez, Esteve Fernández, and Jose M. Martínez-Sánchez. 2017. “Impact of the Spanish Smoking Laws on Tobacco Consumption and Secondhand Smoke Exposure: A Longitudinal Population Study.” *Addictive Behaviors* 75: 30–35. doi: 10.1016/j.addbeh.2017.06.016.
 15. Tassinari, Filippo. 2024. “Low Emission Zones and Traffic Congestion: Evidence from Madrid Central.” *Transportation Research Part A: Policy and Practice* 185: 104099. doi: 10.1016/j.tra.2024.104099.
 16. Abadie, Alberto, and Javier Gardeazabal. 2003. “The Economic Costs of Conflict: A Case Study of the Basque Country.” *American Economic Review* 93(1): 113–132. doi: 10.1257/00028280321455188.
 17. González, Libertad. 2013. “The Effect of a Universal Child Benefit on Conceptions, Abortions, and Early Maternal Labor Supply.” *American Economic Journal: Economic Policy* 5(3): 160–188. doi: 10.1257/pol.5.3.160.